Cover letter

Dear Editors,

Please consider “Limited-label cross-session motor-imagery EEG recalibration: spatial refitting and the added value of Euclidean alignment” as a Paper for Journal of Neural Engineering. The manuscript examines what a short calibration run contributes after a longitudinal motor-imagery decoder has already received the same source and current-session labels. It compares frozen CSP/LDA, classifier-only recalibration, full spatial-filter/classifier refitting and Euclidean alignment followed by full refitting.

EA added 5.30 balanced-accuracy points beyond full refitting in the original twelve-person Kumar cohort and 5.65 points in the nine-person BNCI cohort. Weak-class recall increased by 10.45 and 8.52 points, respectively. Matched-coordinate, RMS and diagonal-reference controls examine the contribution of the alignment operation; prediction transitions identify corrected and introduced errors. The paired Kumar window analysis also establishes a material signal-definition dependence: the added EA contrast was 7.85 points for cue-anchored five-second windows and 0.13 points for matched task-start two-second windows. Participant-level analyses retain these results alongside their uncertainty and longitudinal distribution.

The contribution is an information-matched analysis of recalibration operations, class coverage and signal definitions using established alignment and decoding methods. This combination connects neural signal processing with the practical allocation of short calibration recordings. We provide a complete manuscript PDF with embedded figures and tables, supplementary methods, standalone saved-data CSV files and a code/reproduction guide.

The manuscript addresses a distinct question: what Euclidean alignment adds after supervised spatial-filter and classifier refitting. A separate editor-facing statement identifies the shared Kumar evidence with PONE-D-26-52050, currently under review at PLOS ONE, and the additional alignment, BNCI, class-coverage and controlled-sensitivity analyses.

All three authors have critically reviewed and approved the final manuscript, consent to its submission to Journal of Neural Engineering, and accept accountability for the work. The authors declare no conflict of interest, and this research received no external funding. The manuscript includes the Article 32(1) secondary-analysis exemption basis and the AI-assistance disclosure.

Sincerely,

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